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Some properties of combinatorial triangles related to Horadam polynomials

Yu Wang^a, Jinyang Zhang^a and Huyile Liang^{a,b,c}

^aCollege of Mathematics Science, Inner Mongolia Normal University, Hohhot, People's Republic of China;

^bCenter for Applied Mathematics, Hohhot, People's Republic of China; ^cKey Laboratory of Infinite-dimensional Hamiltonian System and Its Algorithm Application, Ministry of Education, Hohhot, People's Republic of China

ABSTRACT

The Horadam polynomials unify many well-known polynomials, such as the Fibonacci polynomials, the Lucas polynomials, the Pell polynomials, the Jacobsthal polynomials and the Chebyshev polynomials. We investigate some properties of various combinatorial triangles related to Horadam polynomials, including their properties as almost-Riordan arrays and Riordan arrays, their total positivity, the real-rootedness of the generating functions of their rows, and the asymptotic normality (by central and local limit theorems).

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1. Introduction

Horadam [1] introduced in 1965 the sequence h_n having the recurrence relation

$$h_n = ph_{n-1} + qh_{n-2}, \quad n = 2, 3, \dots, \quad (1)$$

with initial conditions $h_0 = a$ and $h_1 = b$. This sequence generalizes many classical sequences, such as the Fibonacci numbers sequence $(F_n)_{n \geq 0}$, the Lucas numbers sequence $(L_n)_{n \geq 0}$, the Pell numbers sequence $(P_n)_{n \geq 0}$, the Pell–Lucas numbers sequence $(Q_n)_{n \geq 0}$, the Jacobsthal numbers sequence $(J_n)_{n \geq 0}$ and the Jacobsthal–Lucas numbers sequence $(j_n)_{n \geq 0}$. These sequences have been studied by many mathematicians for a long time in the field of combinatorial number theory. In addition, they are of great importance in various subjects, such as algebra, geometry, combinatorics, approximation theory, statistics and number theory (see [1–5] for more information).

Horcum and Kocer [2] considered in 2009 the Horadam polynomials $h_n(x)$, which are given by the recurrence relation

$$h_n(x) = pxh_{n-1}(x) + qh_{n-2}(x), \quad n = 2, 3, \dots, \quad (2)$$

CONTACT Huyile Liang  lianghuyile@imnu.edu.cn  College of Mathematics Science, Inner Mongolia Normal University, Hohhot 010022, People's Republic of China Center for Applied Mathematics, Inner Mongolia, Hohhot 010022, People's Republic of China Key Laboratory of Infinite-dimensional Hamiltonian System and Its Algorithm Application, Ministry of Education, Hohhot 010022, People's Republic of China

Table 1. aaa

n	$h_n(x)$
0	a
1	bx
2	$bp^2x^2 + aq$
3	$bp^2x^3 + q(ap + b)x$
4	$bp^3x^4 + pq(ap + 2b)x^2 + aq^2$
5	$bp^4x^5 + p^2q(ap + 3b)x^3 + q^2(2ap + b)x$
6	$bp^5x^6 + p^3q(ap + 4b)x^4 + pq^2(3ap + 3b)x^2 + aq^3$
7	$bp^6x^7 + p^4q(ap + 5b)x^5 + p^2q^2(4ap + 6b)x^3 + q^3(3ap + b)x$
$\vdots$	$\vdots$

The Horadam polynomials $h_n(x)$

with initial conditions $h_0(x) = a$ and $h_1(x) = bx$. The Horadam polynomials share several intriguing properties and connections with other fields of mathematics, such as number theory, combinatorics and algebraic geometry (see also [2,6–8]).

From (2), we list the first few terms of the Horadam polynomials in the following table.

Example 1.1: The Horadam polynomials unify many well-known polynomials, such as

- (i) the Fibonacci polynomials $F_n(x)$, when $a = b = p = q = 1$;
- (ii) the Lucas polynomials $L_{n-1}(x)$, when $a = 2$ and $b = p = q = 1$;
- (iii) the Pell polynomials $P_n(x)$, when $a = q = 2$ and $b = p = 1$;
- (iv) the Pell–Lucas polynomials $Q_{n-1}(x)$, when $a = b = p = 2$ and $q = 1$;
- (v) the Jacobsthal polynomials $J_n(y)$, when $a = b = p = x = 1$ and $q = 2y$;
- (vi) the Jacobsthal–Lucas polynomials $j_{n-1}(y)$, when $a = 2$, $b = p = x = 1$ and $q = 2y$;
- (vii) the Chebyshev polynomials of the first kind $T_{n-1}(x)$, when $a = b = 1$, $p = 2$ and $q = -1$;
- (viii) the Chebyshev polynomials of the second kind $U_{n-1}(x)$, when $a = 1$, $b = p = 2$ and $q = -1$.

We aim to investigate the combinatorial and analytic properties of the triangles related to Horadam polynomials in this paper. By defining $h_n(x) = \sum_{k=0}^n h_{n,k}x^k$, we obtain the numbers $h_{n,k}$, which satisfies the recurrence relation

$$h_{n,k} = ph_{n-1,k-1} + qh_{n-2,k}, \quad n \geq k \geq 0 \quad (3)$$

with initial conditions $h_{0,0} = a$, $h_{1,1} = b$ and $h_{n,k} = 0$ if $n, k < 0$ and $h_{n,k} = 0$ if $n < k$. We call

$$H = [h_{n,k}]_{n,k \geq 0} = \begin{pmatrix} a & & & & \\ 0 & b & & & \\ aq & 0 & bp & & \\ 0 & q(ap + b) & 0 & bp^2 & \\ aq^2 & 0 & pq(ap + 2b) & 0 & bp^3 \\ \vdots & & & & \ddots \end{pmatrix}$$

the *Horadam triangle*, which is derived by arranging the numbers $h_{n,k}$ in a triangular array.

Example 1.2: The Horadam triangle unifies many familiar triangles in [9], such as

- (i) the coefficients of Pell polynomials [9, A115322], when $a = q = 1$ and $b = p = 2$;
- (ii) the coefficients of the Lucas polynomials [9, A114525], when $a = 2$ and $b = p = q = 1$;
- (iii) the coefficients of Chebyshev's $T(n, x)$ polynomials [9, A053120], when $a = b = 1$, $p = 2$ and $q = -1$;
- (iv) the coefficients of Chebyshev's $S(n, x)$ polynomials [9, A049310], when $a = b = p = 1$ and $q = -1$;
- (v) the coefficients of q -Fibonacci polynomials [10] for $F(x, n) = xF(x, n-1) + sF(x, n-2)$, when $a = b = p = 1$ and $q = s$.

We define the compression of the Horadam triangle $H^C = [h_{n,k}^C]_{n,k \geq 0}$ and its reversal $\overleftarrow{H^C} = [\overleftarrow{h}_{n,k}^C]_{n,k \geq 0}$ as follows:

$$\begin{aligned} h_{n,k}^C &:= h_{2n-k,k}, \quad n \geq k \geq 0, \\ \overleftarrow{h}_{n,k}^C &:= h_{n,n-k}^C = h_{n+k,n-k}, \quad n \geq k \geq 0. \end{aligned} \quad (4)$$

The first few rows of H^C and $\overleftarrow{H^C}$ are:

$$H^C = \begin{pmatrix} a & & & & & \\ aq & b & & & & \\ aq^2 & q(ap+b) & bp & & & \\ aq^3 & q^2(2ap+b) & pq(ap+2b) & bp^2 & & \\ aq^4 & q^3(3ap+b) & 3pq^2(ap+b) & p^2q(ap+3b) & bp^3 & \\ \vdots & & & & & \ddots \end{pmatrix}$$

and

$$\overleftarrow{H^C} = \begin{pmatrix} a & & & & & \\ b & aq & & & & \\ bp & q(ap+b) & aq^2 & & & \\ bp^2 & pq(ap+2b) & q^2(2ap+b) & aq^3 & & \\ bp^3 & p^2q(ap+3b) & 3pq^2(ap+b) & q^3(3ap+b) & aq^4 & \\ \vdots & & & & & \ddots \end{pmatrix}.$$

The general term $h_{n,k}^C$ of the compression of the Horadam triangle H^C satisfies the recurrence relation

$$h_{n,k}^C = ph_{n-1,k-1}^C + qh_{n-1,k}^C, \quad n \geq k \geq 0 \quad (5)$$

with initial conditions $h_{0,0}^C = a$, $h_{1,1}^C = b$ and $h_{n,k}^C = 0$ if $n, k < 0$ and $h_{n,k}^C = 0$ if $n < k$.

The general term $\overleftarrow{h}_{n,k}^C$ of the reversal of the compression of the Horadam triangle $\overleftarrow{H}^C$ satisfies the recurrence relation

$$\overleftarrow{h}_{n,k}^C = q\overleftarrow{h}_{n-1,k-1}^C + p\overleftarrow{h}_{n-1,k}^C, \quad n \geq k \geq 0 \quad (6)$$

with initial conditions $\overleftarrow{h}_{0,0}^C = a$, $\overleftarrow{h}_{1,0}^C = b$ and $\overleftarrow{h}_{n,k}^C = 0$ if $n, k < 0$ and $\overleftarrow{h}_{n,k}^C = 0$ if $n < k$.

Example 1.3: (i) If $a = b = p = q = 1$, then $H^C = \overleftarrow{H}^C$ is the Pascal triangle:

$$H^C = \overleftarrow{H}^C = \begin{pmatrix} 1 & & & & & \\ 1 & 1 & & & & \\ 1 & 2 & 1 & & & \\ 1 & 3 & 3 & 1 & & \\ 1 & 4 & 6 & 4 & 1 & \\ \vdots & & & & & \ddots \end{pmatrix}.$$

(ii) If $a = b = q = 1$ and $p = 2$, then H^C is a triangular array of unsigned coefficients of Chebyshev polynomials of the first kind [9, A081277]:

$$H^C = \begin{pmatrix} 1 & & & & & \\ 1 & 1 & & & & \\ 1 & 3 & 2 & & & \\ 1 & 5 & 8 & 4 & & \\ 1 & 7 & 18 & 20 & 8 & \\ \vdots & & & & & \ddots \end{pmatrix}.$$

(iii) If $b = p = q = 1$ and $a = 2$, then $\overleftarrow{H}^C$ is the Lucas triangle [9, A029635]:

$$\overleftarrow{H}^C = \begin{pmatrix} 2 & & & & & \\ 1 & 2 & & & & \\ 1 & 3 & 2 & & & \\ 1 & 4 & 5 & 2 & & \\ 1 & 5 & 9 & 7 & 2 & \\ \vdots & & & & & \ddots \end{pmatrix}.$$

(iv) If $a = q = 1$ and $b = p = m$, where m is a positive integer, then H^C is a triangle of coefficients in the expansion of $(1 + mx)^n$ [9, A013609-A013619]:

$$H^C = \left[\binom{n}{k} m^k \right]_{n,k \geq 0},$$

and $\overleftarrow{H}^C$ is a triangle of coefficients in the expansion of $(m + x)^n$:

$$\overleftarrow{H}^C = \left[\binom{n}{k} m^{n-k} \right]_{n,k \geq 0}.$$

The main objective of this paper is to investigate the combinatorial and analytic properties of the triangles that correspond to the Horadam polynomials. The paper is organized as follows. In Section 2, we show that both the Horadam triangle and the compression of the Horadam triangle are almost-Riordan arrays, and the reversal of the compression of the Horadam triangle is also a Riordan array. In Section 3, we consider the total positivity of the compression of the Horadam triangle and its reversal, and establish a new criterion for the total positivity of almost-Riordan arrays. In Section 4, we consider the roots of the generating functions of their rows, and the asymptotic normality of these triangles.

2. Riordan arrays and almost-Riordan arrays

Riordan arrays play an important unifying role in enumerative combinatorics [11,12]. Let $f(t) = \sum_{n \geq 0} f_n t^n$ and $g(t) = \sum_{n \geq 0} g_n t^n$ be two formal power series. A *proper Riordan array*, denoted by $(g(t), f(t))$, is an infinite lower triangular matrix whose generating function of the k th column is $g(t)f^k(t)$ for $k \geq 0$, where $g_0 \neq 0, f_0 = 0$ and $f_1 \neq 0$. It is well known [13] that the bivariate generating function $R(t, x)$ of the Riordan array $\mathcal{R} = (g(t), f(t))$ is given by

$$R(t, x) = (g(t), f(t)) \frac{1}{1 - xt} = \frac{g(t)}{1 - xf(t)}. \quad (7)$$

The generalization of Riordan arrays has been the subject of several projects and published papers over the last years (see [14–18] for more information). Barry [15] extended the concept of a Riordan array to an almost-Riordan array. The same article defines an *almost-Riordan array* of order 1 as an ordered triple of formal power series $(a(t)|g(t), f(t))$ satisfying $a(t) = \sum_{n \geq 0} a_n t^n$, $g(t) = \sum_{n \geq 0} g_n t^n$ and $f(t) = \sum_{n \geq 0} f_n t^n$, with $a_0 \neq 0, g_0 \neq 0, f_0 = 0$ and $f_1 \neq 0$. We denote the set of almost-Riordan arrays by $\mathcal{R}_a$. The first few rows of an almost-Riordan array $\mathcal{R}_a = (a(t)|g(t), f(t))$ are as follows:

$$\begin{pmatrix} a_0 & & & & & \\ a_1 & g_0 & & & & \\ a_2 & g_1 & g_0 f_1 & & & \\ a_3 & g_2 & g_0 f_2 + g_1 f_1 & g_0 f_1^2 & & \\ a_4 & g_3 & g_0 f_3 + g_1 f_2 + g_2 f_1 & 2g_0 f_1 f_2 + g_1 f_1^2 & g_0 f_1^3 & \\ \vdots & & & & & \ddots \end{pmatrix}.$$

The generating function of the 0th column is given by $a(t)$, while the generating function of the k th column ($k \geq 1$) is given by $tg(t)(f(t))^{k-1}$. The set of almost-Riordan arrays of order 1 forms a group with the multiplication given as below [15]:

$$(a(t)|g(t), f(t)) \cdot (b(t)|u(t), v(t)) = ((a(t)|g(t), f(t))b(t)|g(t)u(f(t)), v(f(t))),$$

where

$$(a(t)|g(t), f(t)) \cdot b(t) = b_0 a(t) + tg(t) \frac{b(f(t)) - b_0}{f(t)}.$$

By construction, the product of two almost-Riordan arrays is still an almost-Riordan array [15].

Theorem 2.1: The Horadam triangle H and the compression of the Horadam triangle H^C can be represented by almost-Riordan arrays as

$$H = \left(\frac{a}{1-qt^2} \middle| \frac{b+(ap-b)qt^2}{(1-qt^2)^2}, \frac{pt}{1-qt^2} \right),$$

$$H^C = \left(\frac{a}{1-qt} \middle| \frac{b+(ap-b)qt}{(1-qt)^2}, \frac{pt}{1-qt} \right).$$

Proof: From the recurrence relation (3), we have $h_{n,0} = qh_{n-2,0}$ with initial conditions $h_{0,0} = a$ and $h_{n,0} = a$ if $n < 0$. Then the generating function of the 0th column of H is $\frac{a}{1-qt^2}$.

Let $h_k(t) = \sum_{n \geq 0} h_{n,k} t^n$ denote the generating function of the k th column ($k \geq 1$) of H . By the recurrence relation (3),

$$h_k(t) = pth_{k-1}(t) + qt^2h_k(t).$$

Thus

$$h_k(t) = \frac{pt}{1-qt^2} h_{k-1}(t).$$

From this recurrence relation and the initial condition,

$$h_1(t) = \sum_{n \geq 0} h_{n,1} t^n = \frac{t}{1-qt^2} \left(b + \frac{apqt^2}{1-qt^2} \right) = t \left(\frac{b+(ap-b)qt^2}{(1-qt^2)^2} \right),$$

and so

$$h_k(t) = t \left(\frac{b+(ap-b)qt^2}{(1-qt^2)^2} \right) \left(\frac{pt}{1-qt^2} \right)^{k-1}.$$

Therefore H can be represented by an almost-Riordan array as

$$H = \left(\frac{a}{1-qt^2} \middle| \frac{b+(ap-b)qt^2}{(1-qt^2)^2}, \frac{pt}{1-qt^2} \right).$$

Similarly, from the recurrence relation (5), we can prove that H^C is also an almost-Riordan array and

$$H^C = \left(\frac{a}{1-qt} \middle| \frac{b+(ap-b)qt}{(1-qt)^2}, \frac{pt}{1-qt} \right).$$

■

Theorem 2.2: The reverse of the compression of the Horadam triangle $\overleftarrow{H^C}$ can be represented as a Riordan array using the following expression:

$$\overleftarrow{H^C} = \left(\frac{a-(ap-b)t}{1-pt}, \frac{qt}{1-pt} \right).$$

Proof: Let $\overleftarrow{h}_k^C(t) = \sum_{n \geq 0} \overleftarrow{h}_{n,k}^C t^n$ denote the generating function of the k th column of $\overleftarrow{H}^C$. By the recurrence relation (6),

$$\overleftarrow{h}_k^C(t) = qt \overleftarrow{h}_{k-1}^C(t) + pt \overleftarrow{h}_k^C(t).$$

Thus

$$\overleftarrow{h}_k^C(t) = \frac{qt}{1-pt} \overleftarrow{h}_{k-1}^C(t).$$

From this recurrence relation and the initial condition

$$\overleftarrow{h}_0^C(t) = \sum_{n \geq 0} \overleftarrow{h}_{n,0}^C t^n = a + \frac{bt}{1-pt} = \frac{a - (ap - b)t}{1-pt},$$

and so

$$\overleftarrow{h}_k^C(t) = \frac{a - (ap - b)t}{1-pt} \left(\frac{qt}{1-pt} \right)^k.$$

■

Corollary 2.3: The bivariate generating function of the Riordan array $\overleftarrow{H}^C$ is given by

$$\overleftarrow{H}^C(t, x) = \frac{a + (b - ap)t}{1 - pt - qtx}.$$

Proof: Combining Theorem 2.2 and the recurrence relation (7), the bivariate generating function of the Riordan array $\overleftarrow{H}^C$ is

$$\overleftarrow{H}^C(t, x) = \left(\frac{a - (ap - b)t}{1-pt}, \frac{qt}{1-pt} \right) \frac{1}{1-xt} = \frac{\frac{a - (ap - b)t}{1-pt}}{1 - x \frac{qt}{1-pt}} = \frac{a + (b - ap)t}{1 - pt - qtx}.$$

■

Corollary 2.4: The row-generating functions $\overleftarrow{h}_n^C(x)$ of the reversal of the compression of the Horadam triangle $\overleftarrow{H}^C$ is

$$\overleftarrow{h}_n^C(x) = \begin{cases} a & \text{if } n = 0; \\ (aqx + b)(p + qx)^{n-1} & \text{if } n \geq 1. \end{cases} \quad (8)$$

Proof: If $n = 0$, it's easy to show that $\overleftarrow{h}_n^C(x) = a$. If $n \geq 1$, by Corollary 2.3,

$$\begin{aligned}\overleftarrow{h}_n^C(x) &= [t^n] \left(\frac{a + (b - ap)t}{1 - pt - qtx} \right) \\ &= [t^n] \left(\frac{a}{1 - pt - qtx} \right) + [t^{n-1}] \left(\frac{b - ap}{1 - pt - qtx} \right) \\ &= [t^n] \left(a \sum_{i=0}^{\infty} (p + qx)^i t^i \right) + [t^{n-1}] \left((b - ap) \sum_{i=0}^{\infty} (p + qx)^i t^i \right) \\ &= (aqx + b)(p + qx)^{n-1}.\end{aligned}$$

■

Theorem 2.5: The general term of the triangle $\overleftarrow{H}^C$ is

$$\overleftarrow{h}_{n,k}^C = \begin{cases} a \cdot 0^k & \text{if } n = 0; \\ \left(ap \binom{n-1}{n-k} + b \binom{n-1}{n-k-1} \right) p^{n-k-1} q^k & \text{if } n \geq 1. \end{cases}$$

Proof: If $n = 0$, it's easy to show that $\overleftarrow{h}_{0,k}^C = a \cdot 0^k$. If $n \geq 1$, from the definition of Riordan array and Theorem 2.2,

$$\begin{aligned}\overleftarrow{h}_{n,k}^C &= [t^n] \frac{a - (ap - b)t}{1 - pt} \left(\frac{qt}{1 - pt} \right)^k \\ &= [t^n] \left(\frac{aq^k t^k}{(1 - pt)^k} + \frac{bq^k t^{k+1}}{(1 - pt)^{k+1}} \right) \\ &= [t^{n-k}] \frac{aq^k}{(1 - pt)^k} + [t^{n-k-1}] \frac{bq^k}{(1 - pt)^{k+1}} \\ &= [t^{n-k}] aq^k \sum_{i=0}^{\infty} \binom{k+i-1}{i} p^i t^i + [t^{n-k-1}] bq^k \sum_{i=0}^{\infty} \binom{k+i}{i} p^i t^i \\ &= \left(ap \binom{n-1}{n-k} + b \binom{n-1}{n-k-1} \right) p^{n-k-1} q^k.\end{aligned}$$

■

3. Total positivity

Following Karlin [19], a (finite or infinite) matrix is called *totally positive* (TP for short) if all its minors are nonnegative. Total positivity theory has been widely studied in both analysis and combinatorics [19–22]. We first review some basic facts about TP matrices. The following lemma is immediate from the classic Cauchy–Binet formula.

Lemma 3.1: The product of two TP matrices is still TP.

Lemma 3.2 ([22, Proposition 1.5]): *The following operations preserve the class of TP matrices.*

- (i) *Multiplying a row (column) by a positive scalar.*
- (ii) *Adding a positive value to the $(0, 0)$ entry of the matrix.*

We define the following Toeplitz matrix

$$[a_{i-j}]_{i,j \geq 0} = \begin{pmatrix} a_0 & & & & \\ a_1 & a_0 & & & \\ a_2 & a_1 & a_0 & & \\ a_3 & a_2 & a_1 & a_0 & \\ \vdots & & & & \ddots \end{pmatrix},$$

where $(a_n)_{n \geq 0}$ is an infinite sequence of nonnegative numbers. We say that the sequence is a *Pólya frequency* (PF for short) sequence if the corresponding Toeplitz matrix is TP. We identify a finite sequence $a_0, a_1, \dots, a_n$ with the infinite sequence $a_0, a_1, \dots, a_n, 0, 0, \dots$. A fundamental characterization for PF sequences is due to Schoenberg and Edrei [23,24], which states that a sequence $(a_n)_{n \geq 0}$ of nonnegative numbers is PF if and only if its generating function has the form

$$\sum_{n \geq 0} a_n t^n = \lambda t^m e^{\gamma t} \frac{\prod_{j \geq 0} (1 + \alpha_j t)}{\prod_{j \geq 0} (1 - \beta_j t)},$$

where $\lambda > 0, m \in \mathbb{N}, \alpha_j, \beta_j, \gamma \geq 0$ and $\sum_{j \geq 0} (\alpha_j + \beta_j) < +\infty$ (see [19, p.412] for instance). In this case, the generating function is called a *Pólya frequency formal power series* (see [20,22] for instance).

In recent years, many researchers focussed on the total positivity of the Riordan arrays (see [25–27] for instance). Chen and Wang [26] gave the following criterion for the total positivity of Riordan arrays (see also [28]).

Lemma 3.3 ([26, Theorem 2.1]): *Let $\mathcal{R} = (g(t), f(t))$ be a Riordan array. If $g(t)$ and $f(t)$ both form Pólya frequency formal power series, then R is TP.*

Mao et al. [27] established the following criterion for the total positivity of Riordan arrays.

Lemma 3.4 ([27, Theorem 3]): *Let $\mathcal{R} = (g(t), f(t))$ be a Riordan array, where $g(t) = \sum_{n \geq 0} g_n t^n$ and $f(t) = \sum_{n \geq 0} f_n t^n$. If the lower Hessenberg matrix*

$$\begin{pmatrix} g_0 & f_0 & 0 & 0 & \cdots \\ g_1 & f_1 & f_0 & 0 & \\ g_2 & f_2 & f_1 & f_0 & \\ \vdots & \vdots & & & \ddots \end{pmatrix}$$

is TP, then so is $\mathcal{R}$.

Now we consider the total positivity of the triangles derived from the Horadam triangle. It is obvious that the Horadam triangle H is not TP, since it has negative minors of order 2. However, the compression of the Horadam triangle H^C and its reversal $\overleftarrow{H^C}$ can be shown to be TP, and furthermore, we establish a new criterion for the total positivity of almost-Riordan arrays.

Let $\mathcal{R}_a = (a(t)|g(t), f(t))$ be an almost-Riordan array. Denote

$$\mathcal{R}_a^+ = \begin{pmatrix} 1 & 0 \\ 0 & \mathcal{R}_a \end{pmatrix} \quad \text{and} \quad \mathcal{R}_a^L = \begin{pmatrix} a_0 & 0 & 0 & 0 & 0 & \cdots \\ a_1 & s_0/a_0 & f_0 & 0 & 0 & \\ a_2 & s_1/a_0 & f_1 & f_0 & 0 & \\ a_3 & s_2/a_0 & f_2 & f_1 & f_0 & \\ \vdots & & & & & \ddots \end{pmatrix},$$

where $s(t) = g(t) - a^+(t)f(t)$ is the generating function of the sequence $(s_n)_{n \geq 0}$, and $a^+(t)$ is the generating function of the backward shifted sequence $(a_1, a_2, \dots)$ of $(a_n)_{n \geq 0}$. We call $\mathcal{R}_a^L$ the *left product matrix* of the almost-Riordan array $\mathcal{R}_a$.

Proposition 3.5: *Let $\mathcal{R}_a^L$ be the left product matrix of the almost-Riordan array $\mathcal{R}_a = (a(t)|g(t), f(t))$. Then*

$$\mathcal{R}_a = \mathcal{R}_a^L \cdot \mathcal{R}_a^+.$$

Proof: Let $\mathcal{R}_a = [r_{n,k}]_{n,k \geq 0}$, in any almost-Riordan array we will consider the blocks

$$\mathcal{R}_a = \left(\begin{array}{c|c} a_0 & 0 \\ \hline \xi & \mathcal{R} \end{array} \right),$$

where $\xi = (a_1, a_2, \dots)^T$ and $\mathcal{R}$ is a Riordan array. Since the generating function of the k th column ($k \geq 1$) of the Riordan array $\mathcal{R}$ is $g(t)f^{k-1}(t)$,

$$\begin{pmatrix} r_{1,k} \\ r_{2,k} \\ r_{3,k} \\ r_{4,k} \\ \vdots \end{pmatrix} = \begin{pmatrix} f_0 & 0 & 0 & 0 & \cdots \\ f_1 & f_0 & 0 & 0 & \\ f_2 & f_1 & f_0 & 0 & \\ f_3 & f_2 & f_1 & f_0 & \\ \vdots & & & & \ddots \end{pmatrix} \begin{pmatrix} r_{1,k-1} \\ r_{2,k-1} \\ r_{3,k-1} \\ r_{4,k-1} \\ \vdots \end{pmatrix}, \quad k = 2, 3, 4, \dots$$

Note that $r_{0,k} = 0$ for $k \geq 1$. Hence,

$$\begin{pmatrix} r_{0,k} \\ r_{1,k} \\ r_{2,k} \\ r_{3,k} \\ \vdots \end{pmatrix} = \begin{pmatrix} * & 0 & 0 & 0 & \cdots \\ * & f_0 & 0 & 0 & \\ * & f_1 & f_0 & 0 & \\ * & f_2 & f_1 & f_0 & \\ \vdots & & & & \ddots \end{pmatrix} \begin{pmatrix} r_{0,k-1} \\ r_{1,k-1} \\ r_{2,k-1} \\ r_{3,k-1} \\ \vdots \end{pmatrix}, \quad k = 2, 3, 4, \dots,$$

where $*$ represents any real number that can be taken at this position. In order to satisfy the initial condition for the above equation, let

$$\begin{pmatrix} r_{0,1} \\ r_{1,1} \\ r_{2,1} \\ r_{3,1} \\ \vdots \end{pmatrix} = \begin{pmatrix} y_0 & 0 & 0 & 0 & \cdots \\ y_1 & f_0 & 0 & 0 & \\ y_2 & f_1 & f_0 & 0 & \\ y_3 & f_2 & f_1 & f_0 & \\ \vdots & & & & \ddots \end{pmatrix} \begin{pmatrix} r_{0,0} \\ r_{1,0} \\ r_{2,0} \\ r_{3,0} \\ \vdots \end{pmatrix}.$$

By solving the above matrix equation,

$$\begin{pmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ \vdots \end{pmatrix} = \begin{pmatrix} r_{0,1}/r_{0,0} \\ (r_{1,1} - f_0 r_{1,0})/r_{0,0} \\ (r_{2,1} - (f_1 r_{1,0} + f_0 r_{2,0}))/r_{0,0} \\ (r_{3,1} - (f_2 r_{1,0} + f_1 r_{2,0} + f_0 r_{3,0}))/r_{0,0} \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 \\ g_0/a_0 \\ (g_1 - a_1 f_1)/a_0 \\ (g_2 - (a_1 f_2 + a_2 f_1))/a_0 \\ (g_3 - (a_3 f_1 + a_2 f_2 + a_1 f_3))/a_0 \\ \vdots \end{pmatrix}.$$

Thus,

$$\begin{pmatrix} r_{0,k} \\ r_{1,k} \\ r_{2,k} \\ r_{3,k} \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots \\ s_0/a_0 & f_0 & 0 & 0 & \\ s_1/a_0 & f_1 & f_0 & 0 & \\ s_2/a_0 & f_2 & f_1 & f_0 & \\ \vdots & & & & \ddots \end{pmatrix} \begin{pmatrix} r_{0,k-1} \\ r_{1,k-1} \\ r_{2,k-1} \\ r_{3,k-1} \\ \vdots \end{pmatrix}, \quad k = 1, 2, 3, \dots$$

It follows that

$$\begin{pmatrix} r_{0,0} & r_{0,1} & r_{0,2} & r_{0,3} & \cdots \\ r_{1,0} & r_{1,1} & r_{1,2} & r_{1,3} & \\ r_{2,0} & r_{2,1} & r_{2,2} & r_{2,3} & \\ r_{3,0} & r_{3,1} & r_{3,2} & r_{3,3} & \\ \vdots & & & & \ddots \end{pmatrix} = \begin{pmatrix} a_0 & 0 & 0 & 0 & 0 & \cdots \\ a_1 & s_0/a_0 & f_0 & 0 & 0 & \\ a_2 & s_1/a_0 & f_1 & f_0 & 0 & \\ a_3 & s_2/a_0 & f_2 & f_1 & f_0 & \\ \vdots & & & & & \ddots \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & r_{0,0} & r_{0,1} & r_{0,2} & \\ 0 & r_{1,0} & r_{1,1} & r_{1,2} & \\ 0 & r_{2,0} & r_{2,1} & r_{2,2} & \\ \vdots & & & & \ddots \end{pmatrix},$$

or equivalently,

$$\mathcal{R}_a = \mathcal{R}_a^L \cdot \mathcal{R}_a^+.$$

■

Theorem 3.6: Let $\mathcal{R}_a = (a(t)|g(t), f(t))$ be an almost-Riordan array. If the left product matrix $\mathcal{R}_a^L$ is TP, then so is $\mathcal{R}_a$.

Proof: Let $\mathcal{R}_a[n]$ be the submatrix consisting of the first $n+1$ columns of $\mathcal{R}_a$. Clearly, any finite submatrix is contained in such a submatrix $\mathcal{R}_a[n]$. So $\mathcal{R}_a$ is TP if and only if all submatrices $\mathcal{R}_a[n]$ are TP for $n \geq 0$. Thus it suffices to show that $\mathcal{R}_a[n]$ is TP for all $n \geq 0$. We proceed by induction on n .

Let $\mathcal{R}_a = [r_{n,k}]_{n,k \geq 0}$. By Proposition 3.5, we have

$$\begin{pmatrix} r_{0,0} & r_{0,1} & \cdots & r_{0,n+1} \\ r_{1,0} & r_{1,1} & \cdots & r_{1,n+1} \\ r_{2,0} & r_{2,1} & \cdots & r_{2,n+1} \\ r_{3,0} & r_{3,1} & \cdots & r_{3,n+1} \\ \vdots & \vdots & & \vdots \end{pmatrix} = \begin{pmatrix} a_0 & 0 & 0 & 0 & \cdots \\ a_1 & s_0/a_0 & f_0 & 0 & \\ a_2 & s_1/a_0 & f_1 & f_0 & \\ a_3 & s_2/a_0 & f_2 & f_1 & \\ \vdots & & & & \ddots \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & r_{0,0} & r_{0,1} & \cdots & r_{0,n+1} \\ 0 & r_{1,0} & r_{1,1} & \cdots & r_{1,n+1} \\ 0 & r_{2,0} & r_{2,1} & \cdots & r_{2,n+1} \\ \vdots & \vdots & & & \vdots \end{pmatrix},$$

or equivalently,

$$\mathcal{R}_a[n+1] = \mathcal{R}_a^L \cdot \begin{pmatrix} 1 & 0 \\ 0 & \mathcal{R}_a[n] \end{pmatrix}. \quad (9)$$

The first matrix $\mathcal{R}_a^L$ on the right-hand side of (9) is TP by assumption, which implies that all a_n, g_n and f_n are nonnegative, and the matrix $\mathcal{R}_a[0]$ is therefore TP. Assume now that the matrix $\mathcal{R}_a[n]$ is TP for $n \geq 0$. Then the second matrix $\begin{pmatrix} 1 & 0 \\ 0 & \mathcal{R}_a[n] \end{pmatrix}$ on the right-hand side of (9) is also TP. Thus the matrix $\mathcal{R}_a[n+1]$ on the left-hand side of (9) is TP from Remark 3.1. The matrix $\mathcal{R}_a$ is therefore TP by induction. ■

Theorem 3.7: If a, b, p, q are all positive, then

- (i) the compression of the Horadam triangle H^C is TP.
- (ii) the reversal of the compression of the Horadam triangle $\overleftarrow{H^C}$ is TP.

Proof: (i) By Theorem 2.1 and Proposition 3.5, the left product matrix of H^C is

$$L(H^C) = \begin{pmatrix} a & & & & \\ aq & b/a & & & \\ aq^2 & bq/a & p & & \\ aq^3 & bq^2/a & pq & p & \\ \vdots & & & & \ddots \end{pmatrix}.$$

From Theorem 3.6, an almost-Riordan array is TP if its left product matrix is TP. Thus it suffices to show that $L(H^C)$ is TP. Let

$$Q = \begin{pmatrix} 1 & & & & \\ q & 1 & & & \\ q^2 & q & 1 & & \\ q^3 & q^2 & q & 1 & \\ \vdots & & & & \ddots \end{pmatrix}, \quad \Lambda_1 = \begin{pmatrix} a & & & & \\ & b/a & & & \\ & & p & & \\ & & & p & \\ & & & & \ddots \end{pmatrix}.$$

Then $L(H^C)$ is obtained from Λ_1 by adding q times each row to the succeeding row. It follows that $L(H^C) = Q\Lambda_1$. Hence it suffices to show that Q and Λ_1 are TP by Lemma 3.1. It's easy to show the diagonal matrix Λ_1 is TP if a, b, p are all positive. The Toeplitz matrix Q can be represented by Riordan arrays as $Q = (\frac{1}{1-qt}, t)$. By Lemma 3.3, the Riordan arrays Q is TP if q is positive. Hence the compression of the Horadam triangle H^C is TP.

(ii) By Lemma 3.4 and Theorem 2.2, the lower Hessenberg matrix of the Riordan array $\overleftarrow{H^C}$ is

$$L(\overleftarrow{H^C}) = \begin{pmatrix} a & & & & \\ b & q & & & \\ bp & pq & q & & \\ bp^2 & p^2q & pq & q & \\ \vdots & & & & \ddots \end{pmatrix}.$$

From Lemma 3.4, a Riordan array is TP if its lower Hessenberg matrix is TP. Thus it suffices to show that $L(\overleftarrow{H^C})$ is TP. Let

$$\Lambda_2 = \begin{pmatrix} a & & & & \\ & q & & & \\ & & q & & \\ & & & q & \\ & & & & \ddots \end{pmatrix}, \quad P_1 = \begin{pmatrix} 1 & & & & \\ b/q & 1 & & & \\ p(b/q) & p & 1 & & \\ p^2(b/q) & p^2 & p & 1 & \\ \vdots & & & & \ddots \end{pmatrix}.$$

Then $L(\overleftarrow{H^C})$ is obtained from P_1 by multiplying the 0th row by a and multiply the k th row ($k \geq 1$) by q . It follows that $L(\overleftarrow{H^C}) = \Lambda_2 P_1$. Hence it suffices to show that Λ_2 and P_1 are TP by Lemma 3.1. By Lemma 3.2 (i) and (ii), if the Toeplitz matrix

$$P = \begin{pmatrix} 1 & & & & \\ p & 1 & & & \\ p^2 & p & 1 & & \\ p^3 & p^2 & p & 1 & \\ \vdots & & & & \ddots \end{pmatrix}$$

is TP, then P_1 is also TP. By applying a similar approach to that of case (i), we can easily show that both P and Λ_2 are TP. ■

Corollary 3.8: *It turns out that the triangles of Example 1.3 are all TP.*

4. Real-rootedness and asymptotic normality

The polynomials $h_n^C(x) = \sum_{k \geq 0} h_{n,k}^C x^k$ and $\overleftarrow{h}_n^C(x) = \sum_{k \geq 0} \overleftarrow{h}_{n,k}^C x^k$ are defined as the row-generating functions of the compression of the Horadam triangle H^C and its reversal $\overleftarrow{H}^C$ respectively. By the recurrence relation (5), we have

$$h_{n+2}^C(x) = (q + px)h_{n+1}^C(x), \quad n \geq 0 \quad (10)$$

with $h_0^C(x) = a$ and $h_1^C(x) = aq + bx$. Let $p(x)$ be a polynomial of degree n with nonzero constant term. Denote by $\overleftarrow{p}(x) := x^n p(1/x)$ the *reciprocal polynomial* of $p(x)$. It is clear that the roots of $\overleftarrow{p}(x)$ are exactly reciprocals of those of $p(x)$. Then we obtain the following remark.

Remark 4.1: The polynomials $h_n^C(x)$ and $\overleftarrow{h}_n^C(x)$ all have only real roots. Furthermore, these polynomials have a root of multiplicity 1 and one of multiplicity $n-1$.

Proof: By the recurrence relations (10), we can obtain the roots of the polynomials

$$h_n^C(x) = \begin{cases} a & \text{if } n = 0; \\ (aq + bx)(q + px)^{n-1} & \text{if } n \geq 1. \end{cases} \quad (11)$$

Therefore, we have the following conclusions. Let $r_{n,i}$ and $\overleftarrow{r}_{n,i}$ be roots of the polynomials $h_n^C(x)$ and $\overleftarrow{h}_n^C(x)$. For $n \geq 1$, the roots of the polynomials $h_n^C(x)$ are

$$r_{n,1} = -aq/b \quad \text{and} \quad r_{n,i} = -q/p, \quad i = 2, 3, \dots, n.$$

Then, the roots of the polynomials $h_n^C(x)$ are real. Note that the polynomials $\overleftarrow{h}_n^C(x)$ are reciprocal polynomials of $h_n^C(x)$. Then the roots of the polynomials $\overleftarrow{h}_n^C(x)$ are

$$\overleftarrow{r}_{n,1} = \frac{1}{r_{n,1}} = -b/aq \quad \text{and} \quad \overleftarrow{r}_{n,i} = \frac{1}{r_{n,i}} = -p/q, \quad i = 2, 3, \dots, n,$$

and the roots of the polynomials $\overleftarrow{h}_n^C(x)$ are real. ■

Let $[a_{n,k}]_{0 \leq k \leq n}$ be a nonnegative matrix and let

$$p(n, k) = \frac{a_{n,k}}{\sum_{j=0}^n a_{n,j}}$$

denotes the normalized probabilities. Following Bender [29], we say that the sequence $a_{n,k}$ is *asymptotically normal* by a *central limit theorem*, if

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}} \left| \sum_{k \leq \mu_n + x\sigma_n} p(n, k) - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt \right| = 0, \quad (12)$$

where μ_n and σ_n^2 are the mean and variance of $(a_{n,k})$, respectively. We say that $a_{n,k}$ is *asymptotically normal by a local limit theorem* on $\mathbb{R}$ if

$$\lim_{n \rightarrow \infty} \sup_{x \in \mathbb{R}} \left| \sigma_n p(n, \lfloor \mu_n + x\sigma_n \rfloor) - \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \right| = 0. \quad (13)$$

In this case,

$$a_{n,k} \sim \frac{e^{-x^2/2} \sum_{j=0}^n a_{n,j}}{\sigma_n \sqrt{2\pi}} \text{ as } n \rightarrow \infty,$$

where $k = \mu_n + x\sigma_n$ and $x = O(1)$. Clearly, the validity of (13) implies that of (12).

Many well-known combinatorial sequences are linked to central and local limit theorems. For example, the famous De Moivre–Laplace Theorem states that the binomial coefficients $\binom{n}{k}$ are asymptotically normal (by central and local limit theorems). Other examples include the signless Stirling numbers of the first kind, the Stirling numbers of the second kind, and the Eulerian numbers (see [30] for instance).

For convenience, we call an infinite lower triangular matrix $[a_{n,k}]_{n,k \geq 0}$ asymptotically normal if the numbers $a_{n,k}$ are asymptotically normal. A standard approach to demonstrating asymptotic normality is the following criterion, which was first used by Harper [31] to show the asymptotic normality of the Stirling numbers of the second kind (see [29, Theorem 2] for instance).

Lemma 4.2: Suppose that $A_n(x) = \sum_{k=0}^n a_{n,k} x^k$ has only real roots, where all $a_{n,k}$ are nonnegative. Let

$$\mu_n = \frac{A'_n(1)}{A_n(1)}, \quad \sigma_n^2 = \frac{A''_n(1)}{A_n(1)} + \mu_n - \mu_n^2. \quad (14)$$

If $\sigma_n^2 \rightarrow +\infty$ as $n \rightarrow +\infty$, then the numbers $a_{n,k}$ are asymptotically normal (by central and local limit theorems) with the mean μ_n and variance σ_n^2 .

Theorem 4.3: If a, b, p, q are all positive, the compression of the Horadam triangle H^C and its reversal $\overleftarrow{H^C}$ are all asymptotically normal.

Proof: By combining (11) and (14) we derive that

$$\begin{aligned} \mu_n &= \frac{(h_n^C)'(1)}{h_n^C(1)} = \frac{np}{p+q} + \frac{q(b-ap)}{(p+q)(aq+b)}, \\ \sigma_n^2 &= \frac{(h_n^C)''(1)}{h_n^C(1)} + \mu_n - \mu_n^2 = \frac{npq}{(p+q)^2} + \frac{q(ap-b)(bp-aq^2)}{(p+q)^2(aq+b)^2}. \end{aligned}$$

Hence, $\sigma_n^2 \rightarrow +\infty$ as $n \rightarrow +\infty$. Therefore, by Theorem 4.1 and Lemma 4.2 the triangle H^C is asymptotically normal. Similarly, by combining (8) and (14), we have

$$\begin{aligned}\overleftarrow{\mu}_n &= \frac{(\overleftarrow{h_n^C})'(1)}{\overleftarrow{h_n^C}(1)} = \frac{nq}{p+q} + \frac{q(ap-b)}{(p+q)(aq+b)}, \\ \overleftarrow{\sigma}_n^2 &= \frac{(\overleftarrow{h_n^C})''(1)}{\overleftarrow{h_n^C}(1)} + \mu_n - \mu_n^2 = \frac{npq}{(p+q)^2} + \frac{q(ap-b)(bp-aq^2)}{(p+q)^2(aq+b)^2}.\end{aligned}$$

Hence, $\overleftarrow{\sigma}_n^2 \rightarrow +\infty$ as $n \rightarrow +\infty$. Therefore, by Theorem 4.1 and Lemma 4.2 the triangle $\overleftarrow{H^C}$ is asymptotically normal. ■

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